EXPERIMENT 14:

ANALYSIS OF EDFA-BASED DENSE WDM LINK PERFORMANCE USING OPTISYSTEM

OBJECTIVES :

1. To design and simulate a Dense Wavelength Division Multiplexing (DWDM) optical communication system using OptiSystem.

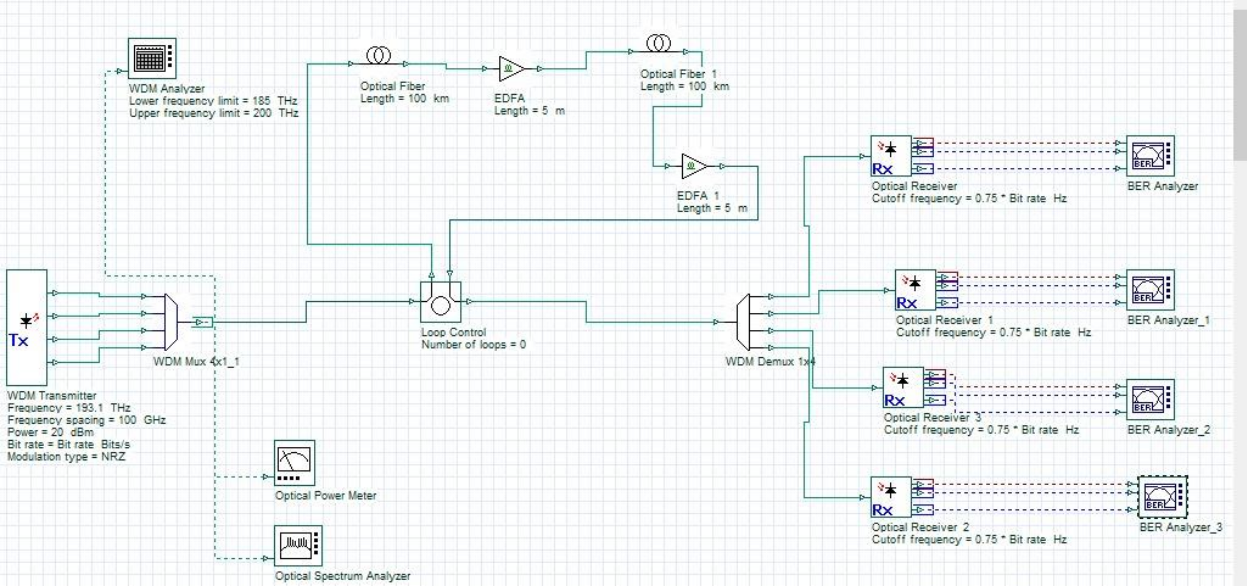
2. To investigate the impact of varying optical fiber length on overall system performance.

3. To examine the effectiveness of an Erbium-Doped Fiber Amplifier (EDFA) in compensating for fiber a enuation losses.

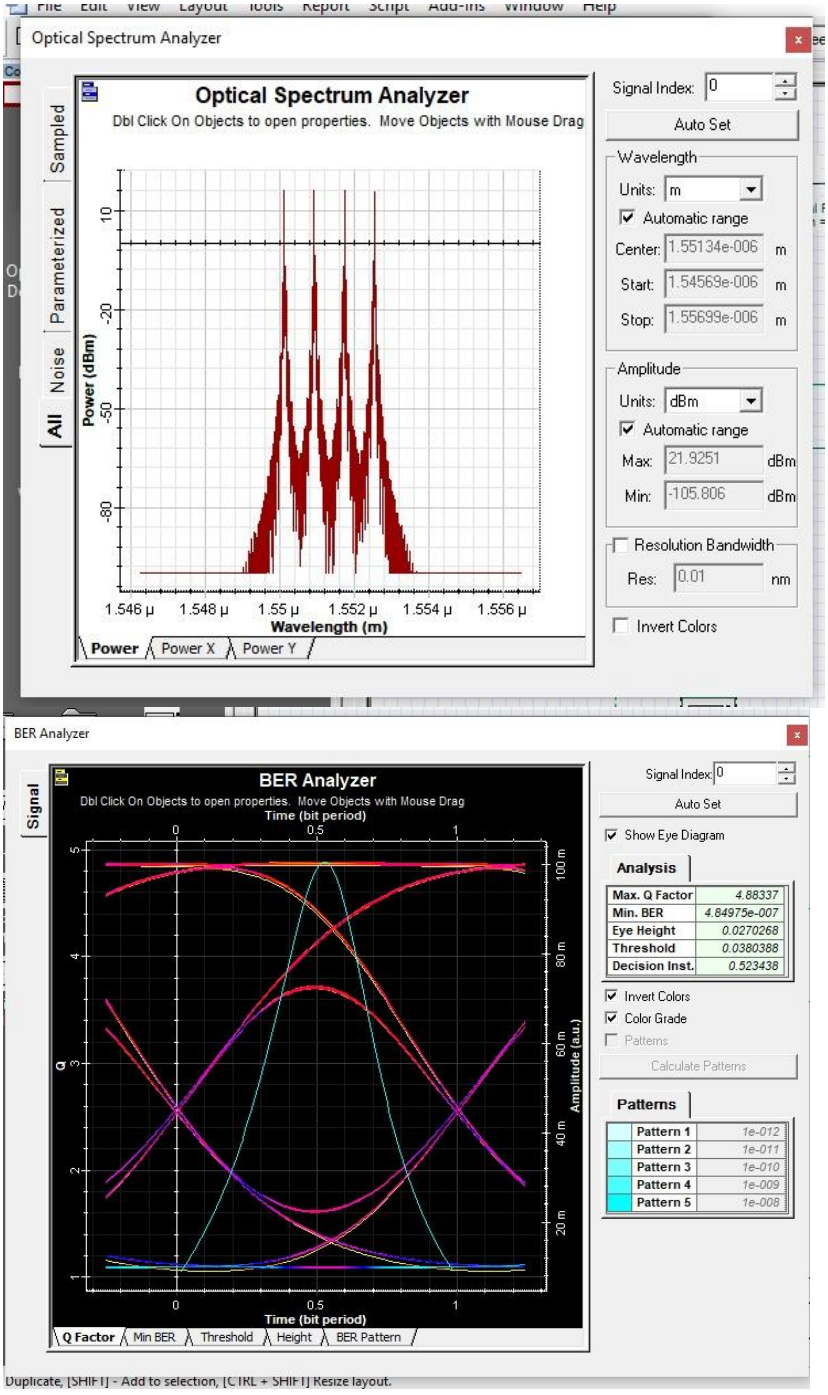
4. To evaluate and compare received power, bit error rate (BER), and Q-factor across different fiber lengths.

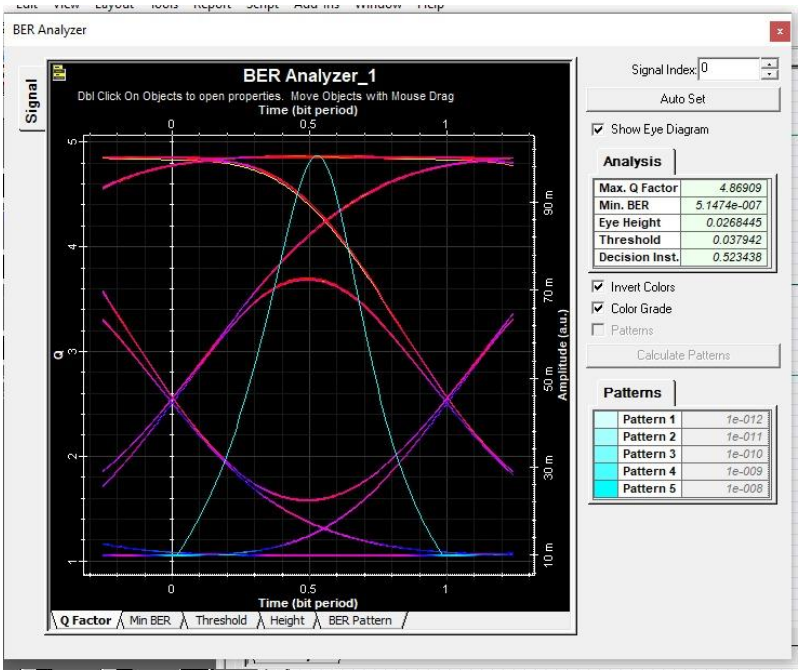
SOFTWARE REQUIRED: Opti System.

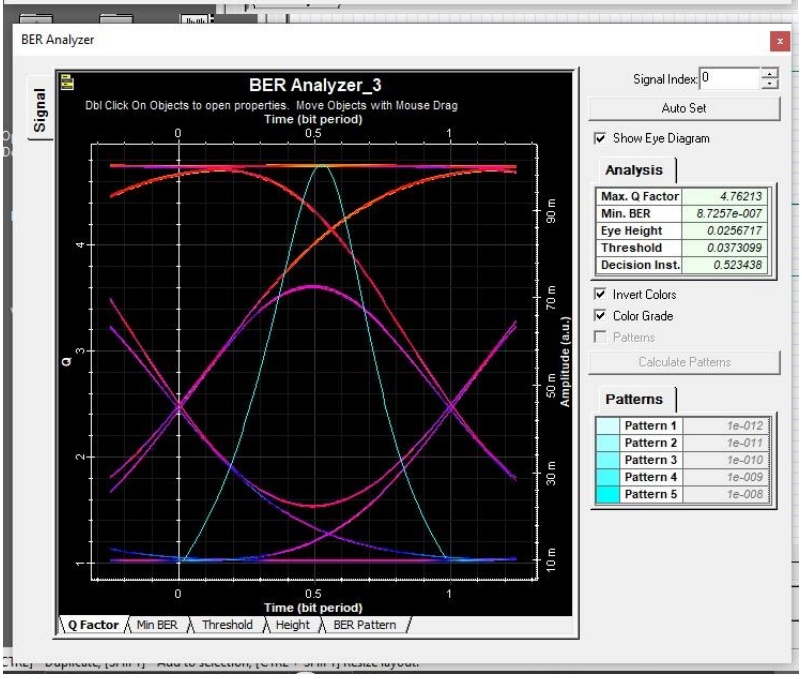
CIRCUIT DIAGRAM:

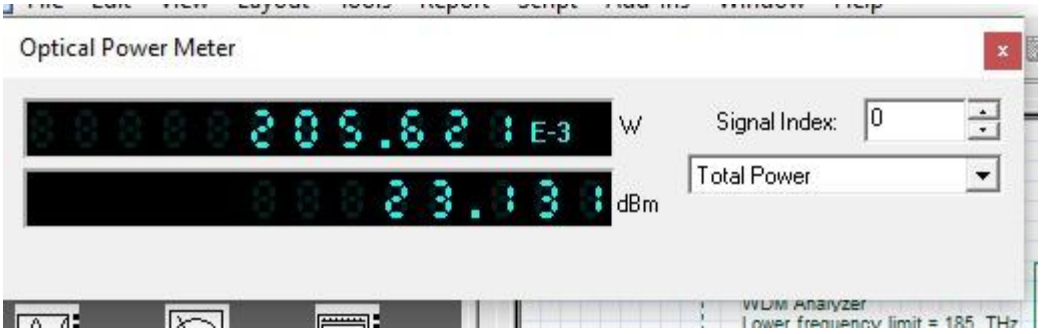


OUTPUTS:


Optical Spectrum Analyzer
Optical Spectrum Analyzer
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Sampled
Parameterized
Noise
All
Power (dBm)
10
-20
-50
-80
1.546 µ
1.548 µ
1.55 µ
1.552 µ
1.554 µ
1.556 µ
Wavelength (m)
Power
Power X
Power Y
Signal Index: 0
Auto Set
Wavelength
Units: m
Automatic range
Center: 1.55134e-006 m
Start: 1.54569e-006 m
Stop: 1.55699e-006 m
Amplitude
Units: dBm
Automatic range
Max: 21.9251 dBm
Min: -105.806 dBm
Resolution Bandwidth
Res: 0.01 nm
Invert Colors
BER Analyzer
BER Analyzer
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Signal
Time (bit period)
Q
Amplitude (a.u.)
100 m
80 m
60 m
40 m
20 m
Q Factor
Min BER
Threshold
Height
BER Pattern
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 4.88337
Min. BER 4.84975e-007
Eye Height 0.0270268
Threshold 0.0380388
Decision Inst. 0.523438
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008
Duplicate, [SHIFT] - Add to selection, [CTRL + SHIFT] Resize layout.


BER Analyzer
BER Analyzer_1
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Signal
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 4.86909
Min. BER 5.1474e-007
Eye Height 0.0268445
Threshold 0.037942
Decision Inst. 0.523438
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008
Q
Amplitude (a.u.)
Q Factor
Min BER
Threshold
Height
BER Pattern


BER Analyzer
BER Analyzer_3
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Signal
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 4.76213
Min. BER 8.7257e-007
Eye Height 0.0256717
Threshold 0.0373099
Decision Inst. 0.523438
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008
Q
Amplitude (a.u.)
Q Factor
Min BER
Threshold
Height
BER Pattern


Optical Power Meter
205.621 E-3
W
23.131
dBm
Signal Index: 0
Total Power

REFLECTION:

learned how multiple optical channels are transmitted through a single fiber using DWDM.

1. I understood the function of an EDFA in compensating for fiber attenuation.
2. I learned to analyze BER and Q-factor using BER analyzers.
3. I understood how fiber length affects optical communication performance

RUBRICS:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improve-ment – 1	Weight-age
Design the OptiSystem PIN/APD receiver circuit with correct components and parameters.	Circuit Setup and Parameter Definition	Builds the complete circuit (laser, modulator, Fork 1×2, attenuators, PIN/APD, filters, power meters, BER analyzers) with all parameters correct (193.1 THz, 0 dBm, 35/38.2 dB, 0.75 × bit rate), independently.	Builds the circuit with minor connection or parameter errors and limited assistance.	Builds only part of the circuit or needs considerable assistance; has unit/parameter errors (e.g., dB vs dBm).	Unable to build the circuit or set the parameters.	25
Run the simulation and record received power, Q factor and BER for both receivers.	Simulation Execution and Data Acquisition	Runs the simulation and records received power (W and dBm), eye diagram, Q factor, BER and the attenuation sweep correctly for PIN and APD.	Records all quantities with minor errors or omissions.	Records some quantities with noticeable errors or inconsistencies.	Unable to run the simulation or record the values.	25
Obtain eye diagrams and BER/SNR plots; determine responsivity, sensitivity and bandwidth.	Graph Plotting and Sensitivity Determination	Plots accurate eye diagrams and BER (semi-log) / SNR curves with labelled axes and a BER = 10^{-9} line; sensitivity within ±1 dB, responsivity and bandwidth within 10% of the reference.	Plots the curves with minor errors; sensitivity, responsivity and bandwidth identified with small deviation.	Plots partial or poorly scaled curves; sensitivity, responsivity or bandwidth approximate.	Unable to plot the curves or determine the parameters.	25
Interpret photocurrent–power relation, SNR/BER variation and PIN vs APD performance.	Analysis and Engineering Interpretation	Correctly explains linear photocurrent, noise-limited regions, SNR/BER trend and the APD gain advantage, and links results to detector selection.	Explains the trends with minor errors and reasonable interpretation.	Performs basic interpretation with noticeable errors or limited justification.	Unable to interpret the results.	25

CONCLUSION :

The DWDM optical communication system was designed and simulated successfully using OptiSystem. The performance of the link was analyzed for different fiber lengths, and the role of the EDFA in compensating for fiber attenuation was studied. The BER and Q-factor results were recorded to evaluate signal quality. The experiment demonstrated the importance of optical amplification in long-distance, high-capacity optical communication systems.